

BUFFALO, MN, USA

WMO: 722114

Lat: 45.159N

Lon: 93.843W

Elev: 967

StdP: 14.19

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54901

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.8	-8.4	-19.8	1.9	-11.9	-15.9	2.4	-7.8	20.9	16.4	18.8	18.8	4.1	280	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	88.4	72.7	85.7	71.1	83.3	69.3	76.0	85.0	73.9	82.8	71.9	80.1	8.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	126.8	82.3	70.9	118.0	80.6	69.0	110.3	77.9	40.1	85.0	38.1	82.8	36.3	79.9	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.5	17.2	15.6	DB	-20.3	93.3	4.7	2.9	-23.7	95.4	-26.5	97.1	-29.2	98.8	-32.6	100.9	(4)
			WB	-20.7	79.1	4.7	2.3	-24.1	80.8	-26.8	82.2	-29.5	83.5	-32.9	85.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	14.9	17.0	31.7	45.2	58.2	68.1	72.6	69.6	62.5	48.0	33.4	19.7	(6)
(7)		DBStd	22.62	13.18	12.03	12.45	10.05	8.89	5.86	5.33	5.02	8.01	8.90	10.59	11.62	(7)
(8)		HDD50	4413	1088	924	578	205	25	0	0	0	5	146	503	939	(8)
(9)		HDD65	7853	1553	1344	1032	596	249	34	6	14	141	533	947	1404	(9)
(10)		CDD50	2671	0	0	12	60	279	542	701	607	380	83	7	0	(10)
(11)		CDD65	636	0	0	1	1	38	126	242	156	66	6	0	0	(11)
(12)		CDH74	5244	0	0	3	21	397	1011	2107	1160	496	49	0	0	(12)
(13)		CDH80	1425	0	0	0	1	120	258	656	269	115	6	0	0	(13)
(14)	Wind	WSAvg	6.3	6.4	6.6	6.9	8.0	7.1	6.0	5.2	4.6	5.8	6.6	6.8	6.4	(14)
(15)	Precipitation	PrecAvg	31.4	0.7	0.7	1.5	2.8	3.7	4.8	3.9	4.3	3.6	2.6	1.6	0.9	(15)
(16)		PrecMax	50.9	2.0	1.7	3.6	8.0	7.7	13.3	10.4	8.5	8.3	6.6	4.7	2.2	(16)
(17)		PrecMin	21.5	0.1	0.1	0.3	0.6	0.8	1.9	1.0	0.7	0.6	0.6	0.1	0.2	(17)
(18)		PrecStd	6.2	0.6	0.4	0.8	1.7	1.7	2.5	2.2	2.1	2.2	1.6	1.2	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.7	51.8	69.3	77.0	90.3	90.4	94.0	89.8	87.9	80.3	66.0	47.6	(19)
(20)		MCWB	37.2	46.9	57.6	58.1	68.3	73.3	77.2	74.2	72.0	64.4	55.0	42.8	(20)	
(21)		2%	DB	38.0	41.6	60.7	70.6	82.8	86.0	89.1	85.8	83.3	73.2	58.7	40.8	(21)
(22)		MCWB	34.8	37.1	51.8	54.5	65.3	69.7	74.4	72.4	69.4	60.6	50.5	36.8	(22)	
(23)		5%	DB	35.2	37.1	54.8	66.4	78.0	82.7	86.1	82.9	80.2	67.5	53.0	37.0	(23)
(24)		MCWB	33.0	34.0	48.5	52.8	62.8	67.6	72.2	69.5	67.7	56.6	46.3	34.7	(24)	
(25)		10%	DB	32.4	33.6	48.3	62.1	73.3	80.0	83.6	80.5	76.1	62.3	48.4	34.4	(25)
(26)		MCWB	30.9	31.1	43.3	50.5	60.1	66.3	70.7	68.3	65.7	53.6	42.9	32.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.6	47.0	61.4	60.4	74.1	76.0	80.1	76.8	74.5	68.5	56.9	45.6	(27)
(28)		MADB	41.8	52.7	68.2	70.9	83.5	86.7	88.8	85.5	83.2	73.7	63.0	47.6	(28)	
(29)		2%	WB	35.2	37.6	54.2	57.4	68.3	73.1	77.2	74.7	71.4	62.0	51.3	37.3	(29)
(30)		MADB	37.7	40.8	59.3	67.3	78.6	82.2	85.7	82.9	80.3	70.7	57.8	39.5	(30)	
(31)		5%	WB	33.2	34.0	48.3	54.6	65.0	70.7	74.7	72.5	69.2	58.2	46.6	34.9	(31)
(32)		MADB	34.8	36.7	54.3	64.0	73.4	79.3	83.7	79.9	77.2	65.6	52.5	36.9	(32)	
(33)		10%	WB	31.0	31.4	43.2	51.4	62.4	68.2	72.5	70.6	67.2	54.4	43.0	32.9	(33)
(34)		MADB	32.5	33.2	48.2	60.2	71.3	76.4	81.4	77.6	75.1	61.2	48.2	34.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	15.4	17.2	16.7	18.4	19.3	18.0	18.5	18.2	18.9	16.8	14.7	13.4	(35)	
(36)		5% DB	MADB	16.8	18.9	22.9	26.2	26.5	21.9	20.5	20.6	22.3	24.6	21.6	14.9	(36)
(37)		MCWBR	14.1	15.1	15.0	14.9	12.3	10.0	9.5	9.9	11.1	13.8	14.3	12.9	(37)	
(38)		5% WB	MADB	16.4	17.5	20.5	22.4	21.7	18.7	18.7	17.8	19.6	20.4	20.5	13.4	(38)
(39)		MCWBR	14.0	14.2	14.3	14.1	11.7	10.5	10.1	10.2	11.4	13.4	14.5	11.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.266	0.278	0.313	0.369	0.403	0.394	0.393	0.401	0.371	0.317	0.287	0.263	(40)	
(41)		taud	2.416	2.373	2.395	2.308	2.267	2.338	2.364	2.334	2.384	2.534	2.513	2.441	(41)	
(42)		Ebn at Noon	273	290	293	282	275	277	276	269	268	273	264	263	(42)	
(43)		Edh at Noon	23	29	32	39	42	39	37	37	33	24	21	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	549	893	1207	1469	1698	1888	2010	1674	1299	824	525	386	(44)	
(45)		RadStd	65	92	111	140	137	142	72	127	93	112	49	46	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page